

IAN SNIDER

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EXPERIENCE

Lawrence Berkeley National Laboratory - *Lab Affiliate*, Berkeley, CA *Aug 2025 - present*

- Graduate researcher in the Bay Area Neutron Group
- Researched experimental techniques for indirect neutron capture cross-section measurement of short-lived nuclei

Lawrence Livermore National Laboratory - *Graduate Intern*, Livermore, CA *June 2025 - present*

- Developed CRISP (CRITICAL Simulation Pipeline) for parallelizing jobs from the radiation transport code: COG
- Test driven Python development included Poetry, CI/CD, SQLite, MPI, CLI, feature branching, and Sphinx
- Used 2 world-class LLNL HPC clusters Dane and Ruby with Slurm sbatch/srun scripts
- CRISP successfully automated the materials compiling and k_{eff} benchmarking of ~3420 ICSBEP critical reactors

Brookhaven National Laboratory - *Nuclear Science Intern*, Upton, NY *Summers 2022 - 2024*

- Developed BRR (Bayesian Resonance Reclassifier) for heavy-nuclei resonance reclassification on NNDC clusters
- Applied Python machine learning using Scikit-learn and Pytorch to classify resonance spin assignments for capture cross-sections of In-115, Pb-206, and U-238
- Used nuclear data and Random Matrix Theory to create synthetic training sets
- Used the neutron transport code OpenMC to perform perturbative sensitivity analyses of the thermal $1/v$ neutron capture cross-sections for critical reactors on Gd-155, 157 and U-235, 238

Truman State University - *Student Astronomy Researcher*, Kirksville, MO *2021 - 2022*

- Calculated trajectories for ~3000 Starlink satellites to optimize telescope viewing plans and developed a GUI
- Researched long-exposure luminosity data corruption due to satellite interference

PROJECTS

WashU Robotics *MATE ROV & MARINER Projects - Mechanical Lead* *2023 - 2025*

- Collaborated with other students to develop an advanced autonomous underwater vehicle (AUV)
- Prototyped hydrodynamic dive control using an IMU and sonar state estimation with a Kalman Filter
- Built a chassis and buoyancy engine

ESE 4481 Autonomous Aerial Vehicle Control - *Student* *Spring 2025*

- Designed and successfully implemented a *cascaded PID/PD* control system on a CrazyFlie 2.1 quadcopter

Society of Physics Students - *Demo Chair* *2020 - 2023*

- Developed 3 new physics demos, performed demos at meetings, and provided volunteer physics tutoring
- Wrote and proctored exams for 2022 & 2023 Science Olympiads (“Crave the Wave” and “Remote Sensing”)

SKILLS

- **Programming:** C, C++, Python, Go, Bash, Lua, MATLAB, Mathematica, Octave, LaTeX, Typst, Vimscript
- **Software/Technical:** Simulink, SolidWorks, OpenMC, COG, Pytorch, Scikit-learn, Slurm, Git, Linux, Arduino
- **Physics/engineering:** Autonomous Aerial Vehicle Control, Classical Mechanics, Electrodynamics, Electronics, Vibrations, Quantum Mechanics, Mathematical Physics, Nuclear Physics, Fluid Mechanics, Solid Mechanics, Acoustics, Materials Science, Thermal Systems, Turbojets, Ramjets
- **Mathematics:** Linear Algebra, ODEs, Computing Structures, Control Systems, Machine Learning, and Optimizations

EDUCATION

Graduate Researcher in Nuclear Engineering - University of California, Berkeley *Aug 2025 - present*

B.S. Mechanical Engineering - Washington University in St. Louis 4.00/4.00 *May 2025*

B.A. Physics - Truman State University 3.89/4.00 *Dec 2024*